

Filtering a Latent Intensity from Three Channels

A primer, with worked examples, on what is actually wrong
with piecewise Gamma–Poisson in-play updating

How to read this

The long cross-domain report is reference material. This is the entry point. It makes **three claims**, gives each one a plain-English gloss, a formal statement, and a worked example with real arithmetic — and then tells you the single experiment that decides whether any of it is worth building.

Every number in the worked examples was computed, not asserted. Where a result cuts against the thesis, it is reported as such (see Example 3).

THE WHOLE DOCUMENT IN SIX LINES

1. Gamma–Poisson conjugate updating is the *exact* filter — for a λ that *does not move*. That is the bug. It never forgets.
2. Your event timestamps identify λ under $\mathbb{P}$. Your odds identify λ under $\mathbb{Q}$. These are different numbers.
3. The odds have *already seen* your events. Fusing them double-counts.

Claim 3 raises a prior question that makes most of the rest moot, and it is cheap to answer. Go to §6.

1 The setup, stripped of context

We track a system on a finite horizon $t \in [0, T]$. A latent, time-varying intensity $\lambda(t) > 0$ governs the arrival of discrete events. We have exactly three sources of information, and no others:

Channel	Symbol	What it is
C0	$\lambda_0 \sim \Gamma(a_0, b_0)$	A static prior from history.
C1	N_t , jumps at $\tau_1, \tau_2, \dots$	Exact event timestamps. A counting process whose intensity <i>is</i> the thing we want.
C2	Z_t	A live market price on the same outcome, from which an implied λ can be inverted.

The object we want is the conditional law $\pi_t(\cdot) = \text{Law}(\lambda_t \mid \mathcal{Y}_t)$, where $\mathcal{Y}_t = \sigma(N_s, Z_s : s \leq t)$ — and, for pricing, the law of the **remaining integrated intensity**

$$\Lambda_{t,T} = \int_t^T \lambda_s ds, \quad \text{because} \quad N_T - N_t \mid \Lambda_{t,T} \sim \text{Poisson}(\Lambda_{t,T}).$$

IN PLAIN ENGLISH

$\Lambda_{t,T}$ is “how many more events do we expect, integrated over the time that is left.” It is the *only* thing that matters for pricing. Not λ at this instant — the integral of λ over the remaining horizon. As $t \rightarrow T$ this goes to zero no matter how large λ is, which is why everything gets strange near expiry.

This is a textbook problem with a name: **nonlinear filtering of a doubly stochastic (Cox) intensity under mixed observation channels**. It is not a football problem. The mature literature on the exact three-channel version is *reduced-form credit risk*, because a credit default swap is precisely a live market price on a latent arrival intensity.

2 Claim 1: your filter is exactly right, for the wrong model

Claim 1. *Gamma–Poisson conjugate updating is the exact, finite-dimensional, closed-form optimal nonlinear filter — for a static Cox intensity. It is optimal and it is degenerate.*

Let $\lambda_t = \lambda_0 \Phi(t)$ with Φ a *known deterministic* modulation (injury-time multipliers, score-state multipliers — anything measurable with respect to the observed history). By Jacod’s formula the likelihood of an observed path is

$$\left. \frac{d\mathbb{P}^\lambda}{d\mathbb{P}^1} \right|_{\mathcal{F}_t} = \exp \left(\int_0^t \log \lambda_s dN_s - \int_0^t (\lambda_s - 1) ds \right) \propto e^{-\lambda_0 \int_0^t \Phi(s) ds} \lambda_0^{N_t} \prod_i \Phi(\tau_i),$$

so that, multiplying by the $\Gamma(a_0, b_0)$ prior,

$$\lambda_0 | \mathcal{Y}_t^N \sim \Gamma \left(\underbrace{a_0 + N_t}_{\text{shape}}, \underbrace{b_0 + \int_0^t \Phi(s) ds}_{\text{rate}} \right). \quad (1)$$

Two things to notice. The known modulation Φ enters *only* through the compensator $\int_0^t \Phi ds$ — the accumulated *exposure* — and never through the count. And the rate parameter only ever *grows*.

IN PLAIN ENGLISH

The rate parameter b is an “amount of evidence” counter. It starts at b_0 (how much history the prior is worth) and only ever increases. So the filter’s willingness to be moved by a new event only ever *decreases*. It has no mechanism for forgetting, because it was derived under the assumption that there is nothing to forget — that λ is a fixed constant we are gradually learning. If λ genuinely drifts during the match, this filter will lag it, and the lag gets worse as the match goes on.

Why this is not an abstract concern

The sensitivity of the posterior mean to one additional event is

$$\frac{\partial}{\partial N_t} \mathbb{E}[\lambda_0 | \mathcal{Y}_t] = \frac{1}{b_0 + \int_0^t \Phi ds}.$$

So the *entire* responsiveness of the filter is set by b_0 : the strength of the prior. And here is the problem.

WORKED EXAMPLE 1 — an honest prior makes in-play updating do nothing

Work in match-units of exposure, so a full match contributes $\int_0^1 \Phi ds = 1$. Take a prior mean of 1.5 goals. Observe a full match in which the team scores 2.

Prior	$\Gamma(a_0, b_0)$	Prior λ	Posterior λ	Move/goal
Honest (38 matches of history)	$\Gamma(57, 38)$	1.500 ± 0.199	1.513 ± 0.197	0.026
Weak (2 matches)	$\Gamma(3, 2)$	1.500 ± 0.866	1.667 ± 0.745	0.333
Zou et al. ($r_1 = E_H(45)$)	$\Gamma(0.75, 0.5)$	1.500 ± 1.732	1.833 ± 1.106	0.667

With a **statistically honest** prior — one whose variance actually reflects a season of data — watching

an entire match moves your estimate of the team’s scoring rate by **0.013 goals**, and shrinks your uncertainty by **1%**. The in-play information is completely swamped. The filter is not wrong; it is telling you the truth: *two goals is almost no evidence against 38 matches of history*.

This is exactly why Zou et al. report that the variance-matched prior gives $r_1 \in (30, 150)$ and “leads to no improvement,” and why they abandon it for $r_1 = E_H(45) \approx 0.75$ — a prior worth *half a match*. That is not a modelling choice. It is a fudge factor, chosen to make the filter move.

THE POINT

The conjugate filter’s inability to respond is not a bug in the arithmetic — it is the *correct* answer to the wrong question. Asking “what is this team’s fixed strength?” correctly returns “the same as before; you have shown me almost no data.” The question you actually want to ask is “**has this team’s strength changed?**” — and that requires a model in which it *can* change.

The fix: discount the evidence counter

Let the state evolve by a multiplicative Beta shock (Smith’s power-steady model; Harvey–Fernandes local-level count model). Gamma conjugacy is preserved *exactly*, and the prediction step is a one-line discounting of the sufficient statistics:

$$\Gamma(a_{t-1}, b_{t-1}) \xrightarrow{\text{predict}} \Gamma(\delta a_{t-1}, \delta b_{t-1}) \xrightarrow{\text{update}} \Gamma(\delta a_{t-1} + \Delta N_t, \delta b_{t-1} + \Delta \int \Phi), \quad \delta \in (0, 1]. \quad (2)$$

The posterior *mean* is preserved by the prediction step ($\delta a / \delta b = a / b$); the *variance* is inflated by $1/\delta$. That is precisely “I still believe the same thing, but I am less sure of it than I was a minute ago.”

IN PLAIN ENGLISH

δ is a forgetting rate. Every minute, the filter throws away a fraction $(1 - \delta)$ of its accumulated evidence. This stops the evidence counter b from growing without bound, so the filter stays responsive for the whole match instead of freezing up. Setting $\delta = 1$ recovers exactly what you have now.

WORKED EXAMPLE 2 — the discount factor derives the number they guessed

The evidence counter now obeys $b_t = \delta b_{t-1} + \Delta$, with $\Delta = 1/90$ per minute. This is a geometric series with a *fixed point*:

$$b^* = \frac{\Delta}{1 - \delta} \quad (\text{the steady-state “effective memory” of the filter}).$$

δ (per minute)	Effective memory b^*	Half-life
0.9800	0.556 matches	34.3 min
0.98519	0.750 matches	46.4 min
0.9900	1.111 matches	69.0 min
1.0000	∞ (never forgets)	∞

Solve $b^* = 0.75$ for δ and you get $\delta = 0.98519$ — whose forgetting half-life is **46.4 minutes**, i.e. almost exactly one half of football.

So Zou et al.’s hand-picked $r_1 = E_H(45)$ is *exactly* the steady state of a discount filter with a one-half half-life. Their fudge factor is a real parameter in disguise. The difference is that δ is (i) principled, (ii) applied *recursively* rather than once at kick-off, and (iii) **estimable from data** — you can fit it by maximising the one-step-ahead predictive likelihood over a historical panel, which you cannot do with a hyperparameter you fixed by hand.

THE POINT

This is a **one-line change** to your existing updater, it preserves conjugacy exactly, it costs two multiplications, and it replaces a guessed constant with a fitted one. If you do nothing else from this document, do this.

3 Claim 2: events and odds measure different things

Claim 2. *The counting channel identifies λ under the physical measure $\mathbb{P}$. The market channel identifies λ under the pricing measure $\mathbb{Q}$. These differ by a multiplicative premium γ , and $\gamma \neq 1$.*

The Girsanov change of measure for a point process multiplies the intensity by a predictable, positive process:

$$\boxed{\lambda_t^{\mathbb{Q}} = \gamma_t \cdot \lambda_t^{\mathbb{P}}} \quad \frac{d\mathbb{Q}}{d\mathbb{P}} \Big|_{\mathcal{F}_t} = \exp \left(\int_0^t \log \gamma_s dN_s - \int_0^t (\gamma_s - 1) \lambda_s^{\mathbb{P}} ds \right). \quad (3)$$

In credit risk γ is the **jump-to-default risk premium**, and it is measured directly: estimate $\lambda^{\mathbb{P}}$ from realised default *events*, estimate $\lambda^{\mathbb{Q}}$ from CDS *prices*, take the ratio. It comes out around $2\times$, it is time-varying, and it is strongly counter-cyclical. In your setting γ is the overround plus whatever systematic distortion the market carries (a favourite-longshot bias is exactly a γ that varies with the level of λ).

IN PLAIN ENGLISH

The market price is not a noisy measurement of the true rate. It is a noisy measurement of the *price* of the true rate — which includes the bookmaker’s margin and whatever risk-aversion or bias the market carries. If you average your event-based estimate with your odds-based estimate as though they were two thermometers reading the same room, you get a number that is systematically off in the same direction for the whole match. That is worse than noise: noise averages out, bias does not, and Kelly staking on a biased estimate loses money in a very specific and repeatable way.

WORKED EXAMPLE 3 — inverting a totals ladder (and a negative result)

Take a totals market: **Over 2.5 @ 1.90, Under 2.5 @ 2.00**. Raw implied probabilities $1/1.90 = 0.5263$ and $1/2.00 = 0.5000$ sum to 1.0263: an overround of 2.63%.

Step 1 — de-vig. Two methods:

Method	$\mathbb{Q}(N \geq 3)$	Implied Λ
Proportional (divide by the book sum)	0.5128	2.7264
Shin (1993), insider-trading model, $z = 0.0263$	0.5132	2.7278
<i>Difference</i>		+0.0014 goals

This is a negative result and I am reporting it as one. On a *tight two-way* book, the choice of de-vig algorithm is worth *one thousandth of a goal*. It does not matter. Do not spend a week on it.

It *does* matter on longshots. On a 1.25 / 6.00 / 13.00 three-way book (overround 4.36%), Shin moves the favourite by +1.41 pp and the 13.0 longshot by −0.82 pp — an 11% *relative* error on the longshot. So: use Shin or a power method on 1X2 and correct-score, and don’t agonise over totals.

Step 2 — the inversion, and its sensitivity. Invert $\mathbb{Q}(N \geq 3) = 0.5128$ under $N \sim \text{Poisson}(\Lambda)$ to get $\Lambda = 2.7264$. Now note the identity

$$\frac{\partial}{\partial \Lambda} \mathbb{P}(N \geq k) = \text{pmf}(k-1; \Lambda), \quad \text{here} \quad \frac{\partial \mathbb{Q}(N \geq 3)}{\partial \Lambda} = \text{pmf}(2; 2.726) = 0.2433.$$

This is the “vega” of the contract — how much the *price* moves per unit of *parameter*. Invert it:

$$\text{a one-tick (0.01) price error} \implies \Delta\Lambda = 0.01/0.2433 = \mathbf{0.041 \text{ goals.}}$$

That is the number that should worry you — not the de-vig method. A single tick of bid–ask noise moves your implied intensity by four hundredths of a goal. This is the ill-posedness of the inverse problem, made concrete.

THE POINT

The vega gives you the observation variance *for free*, and it makes the market channel **self-weighting**:

$$R_t \asymp \left(\frac{\frac{1}{2}(\text{ask} - \text{bid})}{|\partial C / \partial \lambda|} \right)^2 + R^{\text{model}}.$$

As liquidity dries up the spread widens and R_t grows. As the contract becomes insensitive to λ (deep in/out of the money, or $t \rightarrow T$ where $\partial C / \partial \lambda \rightarrow 0$), the denominator collapses and R_t *explodes*. Either way the Kalman gain on the market channel automatically goes to zero. **You never hand-tune a weight for the market signal. The spread-over-sensitivity ratio is the weight.**

One market cannot identify two intensities

Proposition 1 (Rank condition). *If the latent state is $\boldsymbol{\lambda} = (\lambda^{(1)}, \dots, \lambda^{(d)})$ and you observe m market functionals $g_1, \dots, g_m$, local identification requires $\text{rank}(\partial g_i / \partial \lambda^{(j)}) = d$. An aggregate market is a function of $\sum_j \lambda^{(j)}$ alone; its Jacobian has rank 1.*

WORKED EXAMPLE 4 — the totals market is blind to the split

Two completely different systems:

$(\lambda^{(1)}, \lambda^{(2)})$	Sum	$\mathbb{P}(N \geq 3)$	Outcome probabilities (1/X/2)
(2.0, 0.8)	2.8	0.5305	0.654 / 0.205 / 0.141
(1.4, 1.4)	2.8	0.5305	0.374 / 0.253 / 0.374

The totals price is **identical to four decimal places** — it must be, since the sum of independent Poissons is Poisson in the sum. But the home-win probability differs by **28.1 percentage points**.

So: a totals market supplies *exactly one* linear constraint, on the sum. It contains *zero* information about the split. Inverting a totals ladder and claiming to have recovered two team intensities is not an approximation — it is unidentified. You need a second, differently-weighted functional (a spread/handicap market, sensitive to $\lambda^{(1)} - \lambda^{(2)}$) to get rank 2.

THE POINT

Similarly, a *single maturity* identifies only the *level* $\Lambda_{t,T} = \int_t^T \lambda_s ds$, never the shape of λ_s on (t, T) . Any “instantaneous implied intensity” extracted from a single full-match market is an artefact of your own Φ . Recovering the shape needs a calendar spread — a market at an interim maturity — exactly as an instantaneous forward hazard rate requires a CDS *term structure*, not a single quote.

4 Claim 3: the odds have already seen your events

Claim 3. *Optimal filtering requires the observation channels to be conditionally independent given the state. They are not. The market price at time t is a functional of an information set that contains $\{N_s : s \leq t\}$. A naive product-of-likelihoods fusion double-counts the point process.*

This is the formal content of Duffie–Lando (2001): when the market observes the state only through noisy information, the intensity *perceived by the market* is not the structural hazard — it is the conditional hazard given the market’s filtration. **The market’s implied intensity is itself a posterior.** It is not a sensor reading of λ ; it is a noisy reading of *another agent’s filter* of λ , run on a strictly larger information set than yours.

IN PLAIN ENGLISH

You and the bookmaker are both watching the same match. When a goal goes in, you update, and the odds move — because *they* updated too, on the same goal. If you then treat the new odds as independent corroborating evidence, you have counted that goal twice. Your filter becomes more confident than the evidence warrants, and because Kelly stakes scale *inversely with variance*, an over-confident filter systematically overstakes.

WORKED EXAMPLE 5 — what the double-count actually costs you

Suppose the event-only filter gives $\hat{x}^N$ with variance $P_N = 0.09$ (sd 0.30), the market gives $\hat{x}^Z$ with variance $P_Z = 0.04$ (sd 0.20), and their true correlation is $\rho = 0.9$ (high, because they are largely watching the same goals).

The naive Kalman fusion assumes $\rho = 0$, weights by inverse variance ($w_N = 0.308$, $w_Z = 0.692$), and *reports*

$$P_{\text{naive}} = (P_N^{-1} + P_Z^{-1})^{-1} = 0.0277 \quad (\text{sd } 0.166).$$

But the *actual* variance of that same weighted combination, with $\rho = 0.9$, is

$$P_{\text{true}} = w_N^2 P_N + w_Z^2 P_Z + 2w_N w_Z \rho \sqrt{P_N P_Z} = 0.0507 \quad (\text{sd } 0.225).$$

	Variance	sd
What the filter <i>reports</i>	0.0277	0.166
What is <i>true</i>	0.0507	0.225
Variance understated by	83%	
Kelly overstates by	1.83×	

You would bet **nearly twice** the correct fraction of bankroll, on every bet, forever — and you would never see it in your backtest, because the backtest would use the same broken variance.

Three fixes, in increasing order of rigour

1. **Innovation orthogonalisation.** Do not feed the market’s *level* into the filter. Feed its *residual* against your own event-only filter: $\tilde{Z}_t = Z_t - \mathbb{E}[Z_t \mid \mathcal{Y}_t^N]$. Everything the market says that your events already implied is projected out; only its genuinely private information updates the state. Cost: run two filters.
2. **Covariance intersection** (Julier–Uhlmann 1997). The robust fusion rule when the cross-correlation is *unknown*:

$$(P^{\text{CI}})^{-1} = \omega P_N^{-1} + (1 - \omega) P_Z^{-1}, \quad \omega = \arg \min \det P^{\text{CI}}.$$

It is guaranteed *non-divergent for any* correlation, at the price of conservatism. In Example 5 it returns $\omega^* = 0$, $P^{CI} = 0.040$, sd 0.200 – which never understates the truth. Given that our correlation is not merely unknown but *known to be high*, that conservatism is exactly what we want.

3. **Model the market as an agent.** Posit that the market runs its own filter on $\mathcal{G}_t = \mathcal{F}_t^N \vee \sigma(U_{[0,t]})$ with U its private signal, so that $Z_t = c_t \hat{x}_t^N + (1 - c_t) \hat{U}_t + \eta_t$, and *solve for $\hat{U}_t$ before fusing*. This is the most principled route, and it is what §6 estimates.

5 The unified architecture, in one page

Separate what is *known* from what must be *filtered*:

$$\lambda_t^{(j)} = \underbrace{\hat{\lambda}_0^{(j)}}_{\text{static prior (C0)}} \cdot \underbrace{\Phi_j(t, S_{t-})}_{\text{KNOWN modulation: drift} \times \text{epochs} \times \text{state}} \cdot \underbrace{\exp(X_t^{(j)})}_{\text{LATENT deviation (the only filtered part)}} \quad (4)$$

The injury-time multipliers $\rho(t)$ and the score-state multipliers τ_{xy} are *predictable given the observed history*. They belong in Φ , where they cost nothing. Putting them in the filter would cost dimensions for no information. Only X – the log-deviation of the true intensity from its prior – is latent.

State dynamics (Black–Karasinski / exponential-OU, so $\lambda > 0$ automatically):

$$dX_t^{(j)} = \underbrace{-\kappa_j X_t^{(j)} dt}_{\text{mean-revert to the prior}} + \underbrace{\sigma_j(t) dW_t^{(j)}}_{\text{true strength drifts}} + \underbrace{\sum_k \zeta_{j,k} dH_t^{(k)}}_{\text{structural breaks (red cards)}}, \quad d\beta_t = -\kappa_\beta(\beta_t - \bar{\beta})dt + \sigma_\beta dW_t^\beta, \quad (5)$$

where $\beta = \log \gamma$ is the **market bias state** from Claim 2. Note κ is the continuous-time forgetting rate: $\delta \approx e^{-\kappa \Delta}$. It is the same parameter as (2), and it is the single most important number in the system.

The filter (Kushner–Stratonovich with mixed channels):

$$\begin{aligned} d\pi_t(\varphi) &= \pi_t(\mathcal{A}_t \varphi) dt && \text{(i) drift toward prior} \\ &+ \sum_m \left[\pi_t(\varphi g_m) - \pi_t(\varphi) \pi_t(g_m) \right] (R_t^{(m)})^{-1} \omega_t^{(m)} \left(dZ_t^{(m)} - \pi_t(g_m) dt \right) && \text{(ii) market innovation} \\ &+ \sum_j \left[\frac{\pi_{t-}(\varphi \lambda^{(j)})}{\pi_{t-}(\lambda^{(j)})} - \pi_{t-}(\varphi) \right] \left(dN_t^{(j)} - \pi_{t-}(\lambda^{(j)}) dt \right) && \text{(iii) point-process innovation} \end{aligned} \quad (6)$$

IN PLAIN ENGLISH

Read the three lines as three forces acting on your belief:

- **(i)** With no new data, your belief relaxes back toward the pre-match prior and grows more uncertain. The prior is not a one-shot starting value – it is an *attractor*.
- **(ii)** When the price moves, correct in proportion to how surprising the move was, scaled by how noisy the quote is (R_t , from the spread-over-vega) and tempered by ω_t to avoid double-counting (Claim 3).
- **(iii)** When an event fires, multiply your posterior by λ and renormalise (for a Gamma: $a \mapsto a + 1$). When an event *fails* to fire, the compensator term $-\pi(\lambda) dt$ drags your belief *down*, at exactly the rate you expected events to arrive. **No news is news.**

EKF or particle filter?

WORKED EXAMPLE 6 – why the Gaussian approximation fails here specifically

Suppose it is half-time and no goals have been scored. With prior $\Gamma(3, 2)$ the exact posterior is $\Gamma(3, 2.5)$: mean 1.200, sd 0.693, **skewness** 1.155.

The EKF replaces this with $\mathcal{N}(1.200, 0.693^2)$. That Gaussian:

- assigns $\mathbb{P}(\lambda < 0) = 4.2\%$ – to a rate parameter;
- and, far more damagingly, *mis-states the upper tail*:

	True (Γ)	Gaussian (EKF)	Error
$\mathbb{P}(\lambda > 2.5)$	0.0517	0.0303	–41%
$\mathbb{P}(\lambda > 3.0)$	0.0203	0.0047	–77%

And the tail is *the thing you are pricing*: an over/under contract is $\mathbb{P}(N_T - N_t \geq k)$, a tail functional. The EKF is wrong by a factor of four exactly where the money is.

	Continuous-discrete EKF	Sequential Monte Carlo
Cost per step	$O(d^2)$, $d \approx 2$ – negligible	$O(Pd)$, $P \sim 10^3$ – 10^4 – <i>also</i> negligible
Survival term $e^{-\int \lambda}$	Gaussian moment-matching only	Exact (a per-particle path integral)
Low-count skew	Badly wrong (Example 6)	Exact
Tail functionals	Poor – and this is what you price	Exact
Regime jumps in Φ	Needs re-linearisation	Trivial
Degeneracy risk	None	Yes – resample when $\text{ESS} < P/2$

THE POINT

Use SMC. The state dimension is 1–3, the horizon is short, the events number ≤ 10 : a 10^4 -particle bootstrap filter costs microseconds. The *entire* argument for the EKF is computational, and it is worth nothing here – while the argument against it (bias in the low-count, skewed, tail-functional regime) bites maximally. Rao-Blackwellise: conditional on the log-OU path, the level retains Gamma conjugacy, so sample only the dynamic component and integrate the level analytically.

Keep the closed-form discount filter (2) as the *production* path (nanoseconds, no tuning beyond δ), and SMC as the reference against which you measure its approximation error.

6 What to actually do

title=THE ONE EXPERIMENT THAT DECIDES EVERYTHING

Claim 3 raises a prior question: **does the market channel contain anything your event stream does not?** If the market’s implied intensity is just a mechanical restatement of the scoreline, then there is no second channel – there is one channel and a de-vigging model, and the entire fusion architecture is dead weight.

This is cheap to find out and you already have the data.

Over your historical in-play panel, at each observation time t :

1. Run your existing conjugate filter on the **event stream alone** $\Rightarrow$ event-only log-intensity $\hat{x}_t^N$.
2. De-vig the odds and invert the totals ladder $\Rightarrow$ market-implied log-intensity $\hat{x}_t^M$.

3. Regress $\hat{x}_t^M$ on $\hat{x}_t^N$ (and time-to-go). The R^2 measures how much of the market is simply **the scoreline echoed back at you**. Keep the residual u_t .
4. **The real test**. Regress the *realised* remaining goals $N_T - N_t$ on $\hat{x}_t^N$ **and** u_t :

$$(N_T - N_t) = \alpha + \beta_1 \hat{x}_t^N + \beta_2 u_t + \varepsilon.$$

Outcome	What it means
$\beta_2 \approx 0$	The market has <i>no</i> incremental information beyond your events. Kill the fusion architecture. Implement (2), fix the de-vig, and move on.
$\beta_2 \neq 0$	That coefficient is your edge . And it directly calibrates ω_t – the temper in (6) – which you would otherwise have had to guess.

It is a day’s work against the existing DataStore, and it determines whether the rest of this is worth building. **Run it before you write any filter code.**

If it passes, the build order is

1. **Discount filter** (2). Fit δ by maximising one-step-ahead predictive likelihood. One line; strictly dominates what you have.
2. **Move Φ out of the filter** (4). Epoch and score-state multipliers are predictable; they are free.
3. **Self-weighting market channel** via spread-over-vega (R_t). No hand-tuned weights.
4. **Premium state** $\beta = \log \gamma$ (3). Estimate offline first; if it is stable, fix it.
5. **Temper / orthogonalise** the market channel (ω_t from step 4’s β_2).
6. **SMC** as reference; keep the closed-form filter in production.

Sources for the load-bearing claims

All identifiers below were retrieved from OpenAlex/arXiv and checked to resolve.

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- **Segall & Kailath (1975)**, “The modeling of randomly modulated jump processes.” 10.1109/tit.1975.1055359 – the martingale-innovations form of (6).
- **Harvey & Fernandes (1989)**, “Time series models for count or qualitative observations,” *JBES*. 10.1080/07350015.1989.10509750 – **the discount filter** (2).
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- **Duffie & Lando (2001)**, “Term structures of credit spreads with incomplete accounting information,” *Econometrica*. OpenAlex w2118891973 – **the market’s intensity is itself a posterior** (Claim 3).
- **Berndt et al.**, “Measuring default risk premia from default swap rates and EDFs.” 10.2139/ssrn.556080 – direct measurement of the γ wedge (Claim 2).

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- **Cori et al. (2013)**, “A new framework and software to estimate time-varying reproduction numbers during epidemics,” *Am. J. Epidemiol.* 10.1093/aje/kwt133 — the same Gamma–Poisson filter, with a forgetting window, in epidemiology.
- **Brown, Frank & Eden (2004)**, “Dynamic analysis of neural encoding by point process adaptive filtering,” *Neural Computation*. 10.1162/089976604773135069.

Caveat. All derivations here are assembled from the cited frameworks; the specific composition is not itself a retrieved result. The worked examples were computed numerically. A literature search found **no** prior work fusing a historical prior, an event stream, and a concurrent market price of the outcome outside of finance — so the credit-risk literature is the only mature guide to channels C1 and C2 *together*.